

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Intellectual Property Rights

Group Activity Task

Group 1: Copyrights

Group 2: Patents

Group 3: Confidential Information (Trade Secrets)

Group 4: Trademark

Group 5: Authors Guild v. Google, Inc. (2015)

**Group 6: Google Inc. v. Oracle America, Inc.
(2021)**

What is Intellectual Property

- If someone steals your laptop, you lose it; if someone copies your design, you both still have it, but your idea is no longer exclusive.
- Physical (tangible) property: can be touched, owned, and stolen (e.g., bicycle, computer).
- Intellectual property: *intangible creations of the mind* can be copied or shared without depriving the original owner.

Tangible vs Intangible Property

Tangible Property

Physical, can be touched

Protected by theft & damage laws

Example: a car, a building

Intellectual Property

Intangible, exists as information or ideas

Protected by **Intellectual Property Rights (IPR)**

Example: a software code, a music composition

Why IP Needs Special Protection

- Information can be copied without removing it from the original owner.
- IP laws ensure creators benefit from their ideas and inventions.
- Without IP protection → no incentive to innovate.

Evolution of IP Laws

Feature	Paris Convention (1883)	Berne Convention (1886)
Focus	Industrial property (patents, trademarks)	Literary & artistic works (copyrights)
Year	1883	1886
Core Principle	National Treatment, Right of Priority	National Treatment, Automatic Protection
Type of IP	Inventions, designs, trademarks	Books, music, art, films, software

DIFFERENT TYPES OF INTELLECTUAL PROPERTY RIGHTS

- **Copyright** is, as the name suggests, concerned with the right to copy something. It may be a written document, a picture or photograph, a piece of music, a recording, or many other things, including a computer program.
- **Patents** are primarily intended to protect inventions, by giving inventors a monopoly on exploiting their inventions for a certain period.
- **Confidential information** is information that a person receives in circumstances that make it clear they must not pass it on.
- **Trademarks** identify the product of a particular manufacturer or supplier.

Copyright

Definition: Right to control copying, distribution, and modification of original works — including software.

Applies to: Source code, object code, manuals, graphics, music, etc.

Example: The Windows operating system's code is protected by copyright, copying or distributing it without permission is illegal.

Patents

Definition: Grants inventors exclusive rights to make, use, or sell an invention for a fixed period (usually 20 years).

Applies to: New inventions, processes, or technical solutions (including some types of software or hardware).

Example: Amazon's “1-Click” purchase system was patented, it simplified online buying.

Confidential Information (Trade Secrets)

Definition: Information shared in confidence that cannot be disclosed or used without permission.

Applies to: Proprietary algorithms, data structures, customer lists, formulas, or strategies.

Example:

The *Coca-Cola formula* has never been patented, it's protected as a trade secret for over 130 years.

Trademarks

Definition: Signs, words, or logos that identify a product or company's brand.

Applies to: Product names, logos, slogans, sounds, or even colors.

Example: The Nike “Swoosh”, Google logo, and Intel jingle are all trademarks.

Comparison

Type	Protects	Duration	Registration Required?	Example
Copyright	Creative works (books, art, code)	Life + 70 years	No (but recommended)	<i>Harry Potter</i> books
Patent	Inventions and processes	20 years	Yes	iPhone multi-touch technology
Trade Secret	Confidential business info	Indefinite (if secret)	No	Coca-Cola recipe
Trademark	Brand identity (logos, names)	Indefinite (renewable)	No (but recommended)	Nike swoosh and slogan

Copyright

- Copyright is associated primarily with the right to copy something. The 'something' is known as the work.
- Only certain types of work are protected by copyright law. The types that concern us here are 'original literary, dramatic, musical or artistic' works.
- Copyright comes into existence when the work is written down or recorded in some other way. It is not necessary to register it in any way.

The rights of the copyright owner

Copyright law gives the owner of the copyright certain exclusive rights. The rights that are relevant to software and, more generally, to written documents, are the following:

- The right to make copies of the work: Making a copy of a work includes copying code from a disc into RAM (random access memory) in order to execute the code.
- The right to issue copies of the work to the public, whether or not they are charged for it.
- The right to adapt the work: This includes translating it whether from English to Chinese or from C to Java.

Copyright Case Studies

- **Authors Guild v. Google, Inc. (2015):** The Authors Guild sued Google over its Google Books project, which scanned millions of books and made portions of them searchable online. Google argued that its use was fair use, as it was transforming the content to create a searchable database, which would help users discover books.
- **Google Inc. v. Oracle America, Inc. (2021):** Oracle sued Google, claiming that Google had copied Java's API (Application Programming Interface) code when developing the Android operating system. Oracle argued that Java's API was copyrighted and that Google's use of it without a license was infringement. Google, on the other hand, argued that its use was fair use, as it involved using the Java code to create a new platform that was transformative.

Databases

- Copyright subsists in a database if its contents constitute the author's own intellectual creation.
- In order to encourage the production of such modest but useful databases, regulations were introduced in 1997 to create a special intellectual property right called the database right.
- The database right subsists in a database 'if there has been substantial investment in obtaining, verifying or presenting the contents of the database'.
- It lasts for 15 years and prevents anyone from extracting or reusing all, or a substantial part of, the database without the owner's permission.

Copyright Infringement

- Infringement occurs when someone uses, copies, or distributes a copyrighted work without the owner's permission.
- Applies to software, text, images, music, videos, etc.
- **Example:** Copying source code from a licensed software into your own app = copyright infringement.

Types of Infringement

Type

Description

Primary Infringement

Directly breaches exclusive rights of the copyright owner (e.g., copying code, distributing software)

Secondary Infringement

Occurs in a *business or commercial context* (e.g., trading or using pirated software)

Example:

A company using unlicensed copies of Microsoft Office across departments = **secondary infringement**.

Case Study – Cantor Fitzgerald vs. Tradition (1999)

Cantor Fitzgerald (CF): financial firm with proprietary bond-broking software.

Michael Howard and **Christopher Harland** left CF, joined competitor **Tradition (UK)**.

They developed a similar, improved system within 3 months